

SHREEVIKAS JAGADISH

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SUMMARY

AI/ML Engineer with experience building **production ML, agentic AI, RAG, predictive modeling, and data systems**. Skilled in **Python, PyTorch, LangGraph, FastAPI, Spark, AWS, and MLOps**, with hands-on experience taking AI products from 0→1, deploying production systems, and iterating features based on **customer and user feedback**. Passionate about building in the AI space, with experience across **customer-facing projects, LLM evaluation, data engineering, and production deployment**.

PROFESSIONAL WORK EXPERIENCE

AI Engineer Intern | *NeuralSeek, USA*

July 2025 – November 2025

- Reduced IT/HR support resolution time by **40%** by conducting **user interviews with non-technical HR/IT stakeholders**, translating requirements into an **enterprise RAG system** using open-source embeddings, PostgreSQL + pgvector, and metadata-filtered retrieval.
- Improved **concurrent LLM serving with vLLM**, tuning **retrieval thresholds, context windows, token budgets, and generation parameters** to increase throughput and reduce unnecessary token usage.
- Reduced **manual compliance-analysis effort by 35%** by **fine-tuning an open-source LLM on AWS with PEFT/QLoRA** for structured, grounded responses with retrieval context, citations, and validation.

Data Scientist (AI/ML) | *Whiterock, India*

Feb 2022 – July 2024

- Improved manufacturing defect-detection accuracy by **22%** while maintaining **98.8% pipeline uptime** by deploying **Scikit-learn and PySpark ML workflows on AWS SageMaker**, with CloudWatch monitoring.
- Partnered with **manufacturing and operations stakeholders to gather requirements**, then developed **time-series forecasting and predictive-maintenance models** using LightGBM, PCA, and anomaly detection to analyze demand and equipment telemetry.
- Built a **BERT + spaCy NLP sentiment pipeline** processing **20K+ customer reviews**, achieving **~89% F1** across positive, neutral, and negative classes while extracting recurring product themes from unstructured feedback.
- Reduced data-preparation time by **45%** and improved reporting performance by **40% for 200+ users** by operating **12 AWS-to-Snowflake ELT pipelines** using Airflow and dbt.

PROJECTS

ArchPilot — Agentic AI System Architecture Copilot (*Next.js, TypeScript, FastAPI, PydanticAI, PostgreSQL, llama.cpp, Local LLMs*)

- Built a **multi-agent system-design copilot** that converts application requirements into **cost-effective, balanced, and high-performance architectures**, using deep web research on current tools and technologies to recommend the optimal design.
- Engineered a **FastAPI + PydanticAI orchestration layer** with **JEV System One decision model for tool/agent routing**, PostgreSQL persistence, and GPU-accelerated local inference using a **27B quantized LLM via llama.cpp**.

AgentShield — Multi-LLM Agent Evaluation & Red-Team Platform (*Next.js, TypeScript, Groq, Gemini, Supabase, Prisma, Zod*)

- Built and deployed a **multi-LLM QA and red-team platform** for AI agents across 6 failure modes and 3 scan levels, integrating Next.js APIs, PostgreSQL persistence, and interactive evaluation dashboards.
- Built a multi-LLM red-team platform testing agents against **prompt injection, privacy leakage, unsafe tool use, hallucinations, policy violations, and excessive-agency failures**, with automated adversarial generation, Gemini-based judging, and regression testing.

AI/ML Knowledge Assistant — Production RAG Application (*Next.js, TypeScript, PostgreSQL/pgvector, Jina, Groq, Vercel*)

- Architected a cloud-native RAG application with pgvector retrieval, Jina embeddings, Groq generation, clickable citations, and similarity-based refusal handling.
- Implemented **exact, semantic, and embedding caching** with latency observability, evolving the system from local FAISS to a lightweight serverless production architecture.

EDUCATION

Master of Science in Information Technology | *Illinois Institute of Technology* | *USA*

May 2026

Bachelor of Computer Science | *Visvesvaraya Technological University* | *India*

August 2023

SKILLS

Programming: Python, SQL, TypeScript, JavaScript, Java, Bash

Machine Learning & Data Science: Scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Classification, Regression, Time-Series Forecasting, Anomaly Detection, PCA, Feature Engineering, Hyperparameter Tuning, A/B Testing

Generative AI & Agentic Systems: LLMs, RAG, Agentic AI, LangGraph, LangChain, LlamaIndex, PydanticAI, OpenAI SDK, JEV, Tool Calling, Prompt Engineering, LLM Evaluation, PEFT, LoRA/QLoRA

Retrieval & LLM Infrastructure: PostgreSQL, pgvector, Embeddings, Semantic Search, Metadata Filtering, vLLM, llama.cpp, Context/Token Optimization, Local LLM Inference

Backend & Data Engineering: FastAPI, REST APIs, Pydantic, Async Python, PySpark, Databricks, Snowflake, dbt, Airflow, Kafka

Cloud & MLOps: AWS, SageMaker, EC2, S3, Lambda, Docker, Kubernetes, MLflow, CI/CD, Terraform, CloudWatch, LangSmith

Frontend & Testing: React, Next.js, Tailwind CSS, Zod, Pytest, Vitest, Playwright

CERTIFICATIONS & RESEARCH

- Graduate Research Assistant:** Researched **Physics-Informed Neural Networks (PINNs)** for power-system dynamics using **PyTorch, NVIDIA PhysicsNeMo, and CUDA-accelerated workflows** for simulation and time-series modeling.
- AWS Certified Data Engineer – Associate (DEA-C01)** — Amazon Web Services
- AI Fluency: Framework & Foundations** — Anthropic
- Google Data Analytics Professional Certificate** — Google